

Model Selection & XAI Guidelines

A Structured Framework for Building Interpretable, High-Performance
Machine Learning Workflows.

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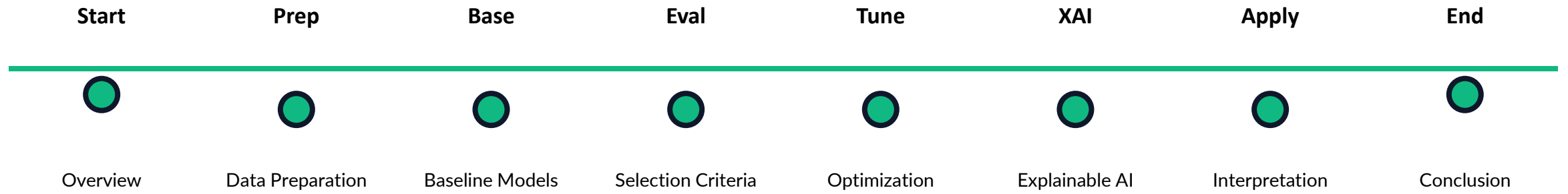
SESSION OVERVIEW

Core Objectives

This interactive session equips you with the end-to-end workflow of **robust model selection** and **Explainable AI (XAI)**. We move beyond "Black Box" predictions to strategic, justifiable insights.

- Master data quality & prep for model selection.
- Establish baseline benchmarks for accuracy.
- Address bias-variance tradeoffs strategically.
- Apply SHAP & LIME for executive reporting.

THE LEARNING ROADMAP



Introduction



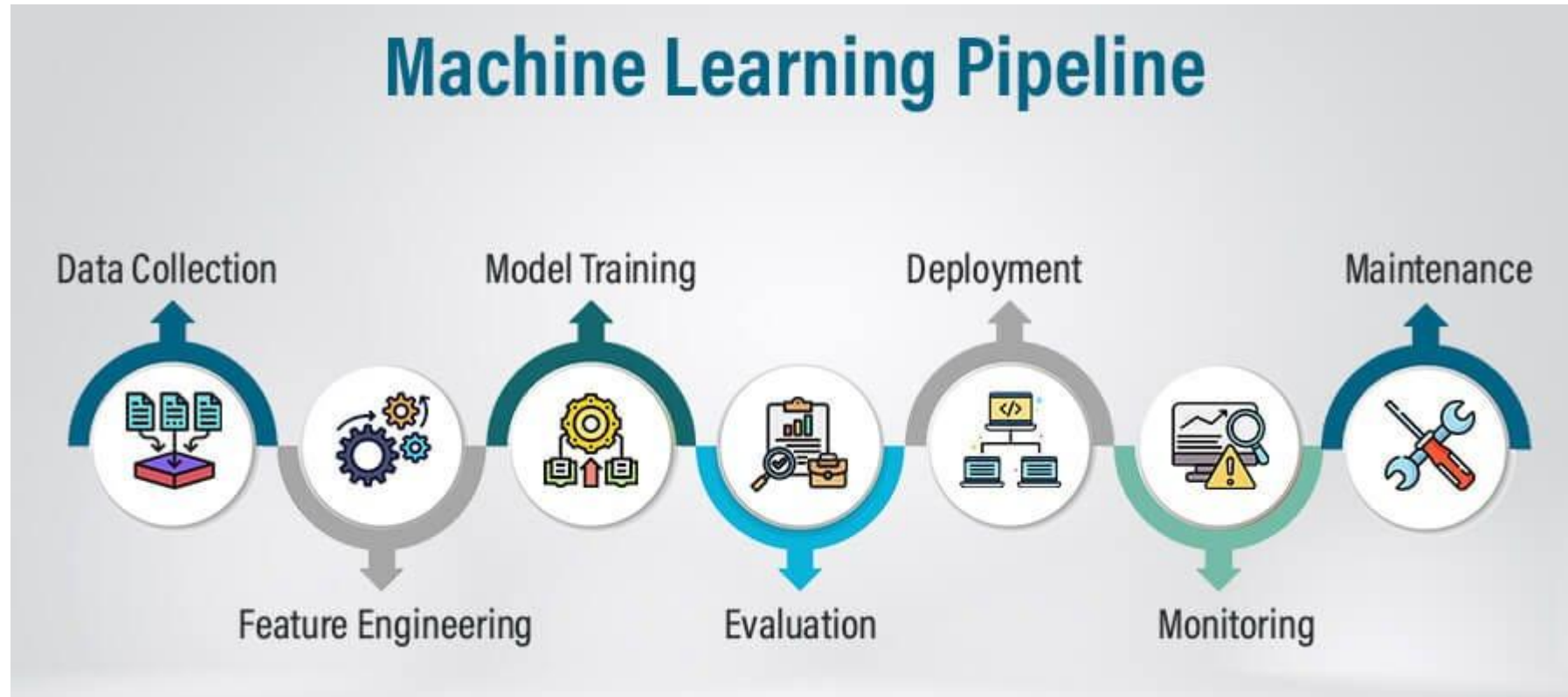
Various Data Sources & Data Types

- Sensor data
- Metrics
- Tickets
- Documents
- Manuals
- Alerts
- Errors
- Logs
- Webcam Images, Videos
- Conversations
- ECG signals, Voltage signals

- Numerical
- Categorical
- Text
- Image
- Video
- Audio
- Signals

Type of Data	Health care Data
Numerical	Blood reports
Categorical	Medical test results
Text	Medical reports
Image	X-ray
Video	CT, MRI, Ultrasound output
Speech	Doctors and technicians discussions and instructions
Signals	ECG, EMG signals

ML Pipeline



Data Quality & Understanding

Before the first model is trained, the outcome is decided by the data. Strategic preparation is the differentiator between noise and signal.

STRATEGIC DATA EXPLORATION



Structure & Types

Analyzing feature distribution, data types, and basic statistics to detect drift or outliers early in the pipeline.



Quality Checks

Identifying missing values, duplicates, and feature relationships that could cause multicollinearity.



Class Balance

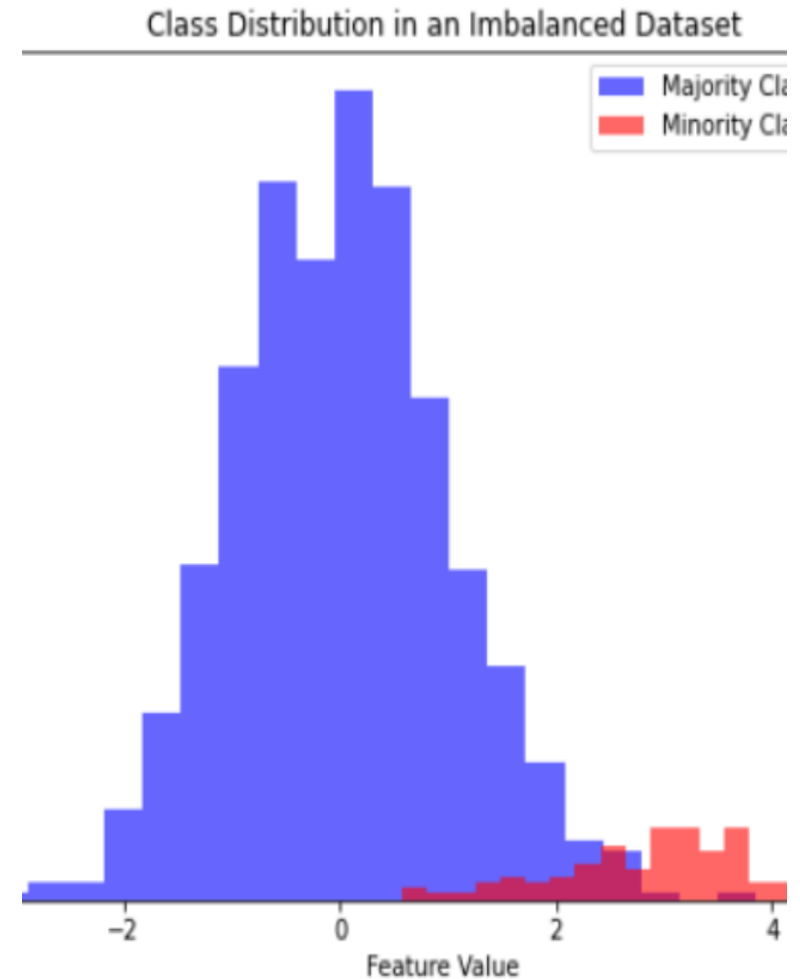
Critical assessment of target variable distribution to prevent algorithmic bias toward majority classes.

HANDLING IMBALANCE

In many real-world cases (Fraud, Rare Diseases), the signal is buried in a tiny fraction of data. We must handle this via:

- **Oversampling (SMOTE):** Synthesizing minority cases.
- **Undersampling:** Reducing majority class dominance.
- **Cost-Sensitive Learning:** Penalizing majority errors.

Goal: Reliable model evaluation on the "rare but critical" cases.



DATA QUALITY DASHBOARD

Check Category	Executive Concern	Actionable Technique
Completeness	Do we have enough signal?	Imputation (Mean/Median) or Deletion
Consistency	Is the data format standardized?	Regex Parsing & Encoding Maps
Class Bias	Will we ignore minority segments?	Stratified Splitting & Weighting
Correlation	Are features redundant?	Heatmap Analysis & VIF Checks

Baseline Model Building

Establishing a "Simple Truth" benchmarks for your data. Before optimizing, we must define what 'Success' looks like without complexity.

MODEL CANDIDATE GALLERY

Linear Baselines

Logistic Regression: High interpretability, assumes linear decision boundaries. Great for "sanity checks".

Tree-Based

Decision Trees: Captures non-linear interactions. Prone to overfitting without pruning.

Ensemble Power

Random Forest: Combines multiple trees to reduce variance and improve stability.

| DEFINING INITIAL SUCCESS

Standard metrics provide the initial pulse of the baseline models before we dive into performance stability and generalizability.

0.82

Accuracy Target

0.78

F1-Score Base

0.85

AUC-ROC Goal

Model Evaluation & Stability

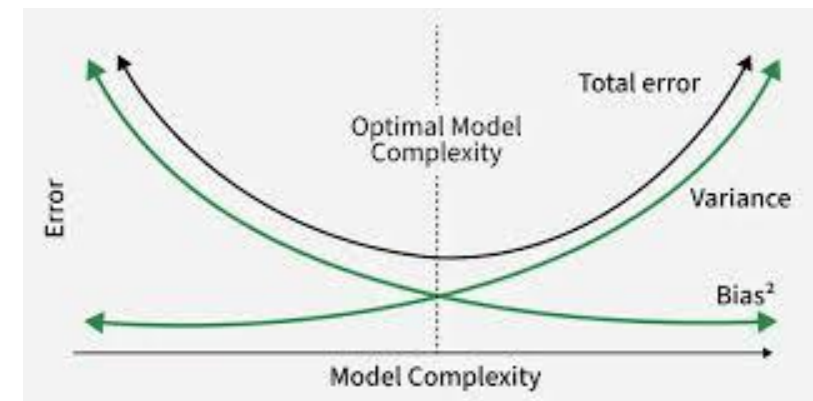
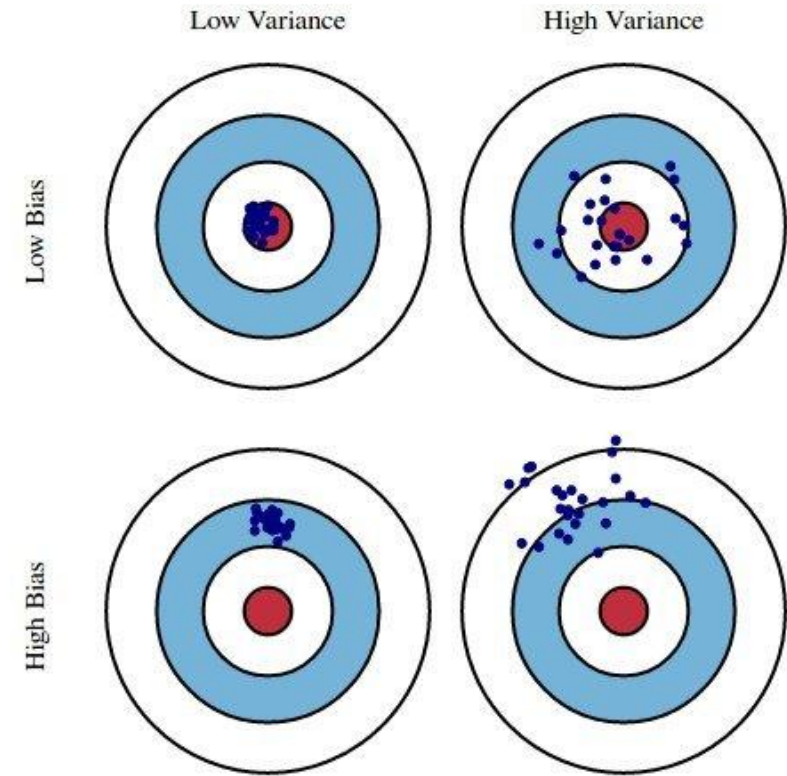
The transition from a 'Model that Works' to a 'Model that Wins'. We look for consistency across unseen data.

BIAS-VARIANCE TRADEOFF

The fundamental challenge in Machine Learning: finding the "Goldilocks" model.

- **High Bias:** Underfitting. The model is too simple to see the patterns.
- **High Variance:** Overfitting. The model memorizes noise instead of learning signal.

We use **Learning Curves** to diagnose where our model sits on this spectrum.



VALIDATION FOR RELIABILITY

Approach	Best For	Risk
Hold-out Split	Large Datasets	Inconsistent samples
K-Fold CV	Medium Datasets	Compute heavy
Stratified CV	Imbalanced Data	Complex setup

Stability Assessment

We look for consistency between training and validation errors. Large gaps signify a failure to generalize, requiring further regularization or feature pruning.

DEEP INSIGHT: CONFUSION MATRIX

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Understanding Errors

Accuracy isn't enough. We must distinguish between types of failure:

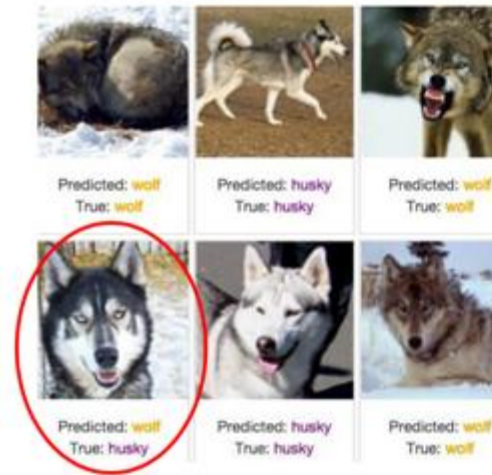
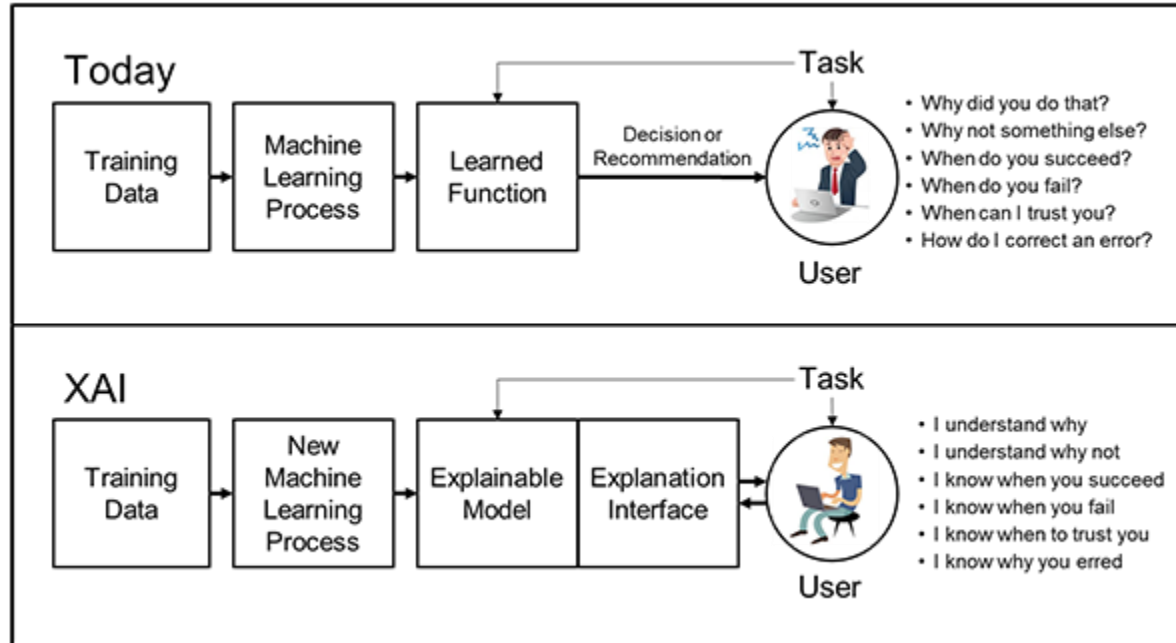
- **False Positives (Type I):** False alarms. Cost of nuisance.
- **False Negatives (Type II):** Missed signals. Strategic risk.

"In Fraud detection, Type II errors are far more expensive than Type I."

Explainable AI (XAI)

The "Why" behind the "What". Turning algorithmic complexity into human-readable strategic justification.

Why xAI?



SHAP VS. LIME COMPARISON

SHAP

Global & Local: Mathematically grounded in Game Theory. Guaranteed consistency across all feature combinations.

SHAP is a method that enables a fast computation of Shapley values and can be used to explain the prediction of an instance x by computing the contribution (Shapley value) of each feature to the prediction. We get contrastive explanations that compare the prediction with the average prediction. The fast computation makes it possible to compute the many Shapley values needed for the global model interpretations.

Best for: Audit & Legal compliance.

LIME

Local Focus: Builds a simple "surrogate" model around a specific prediction to explain individual cases quickly.

“Why is the model behaving in a specific way in the locality of a data point x ?”. LIME only looks at the local structure and not the global structure of the data.

Best for: Fast, intuitive case debugging.

VISUALIZING LIME

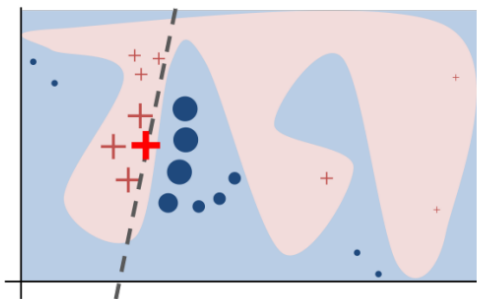
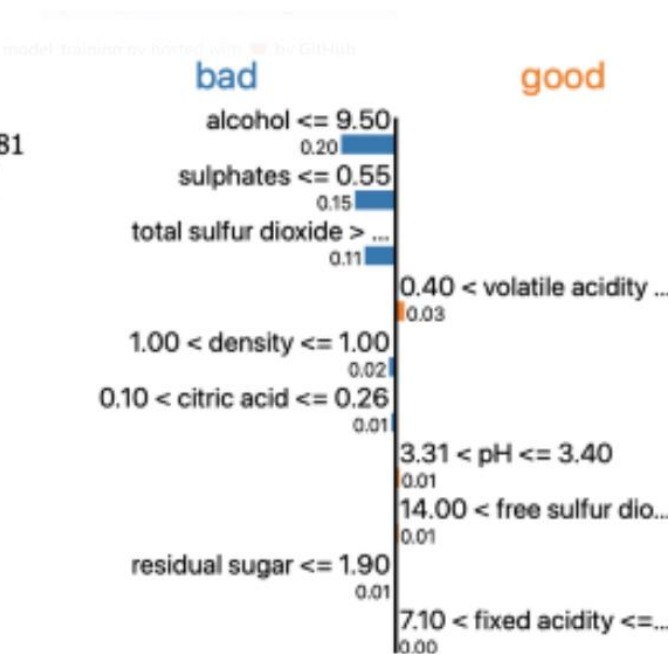


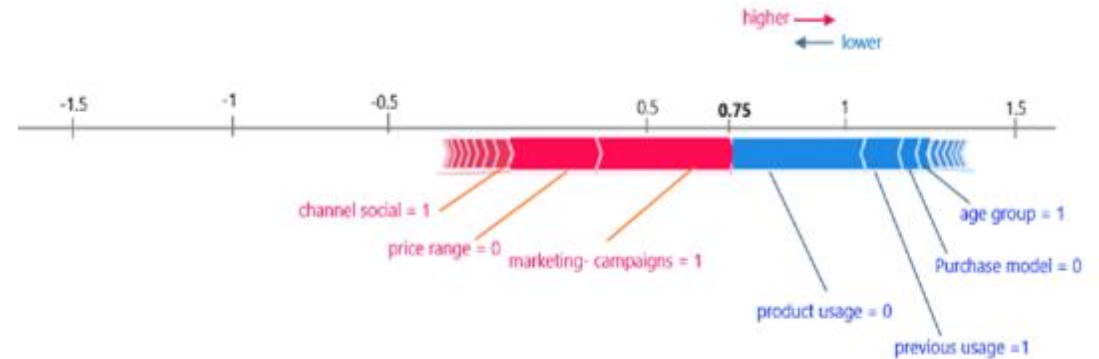
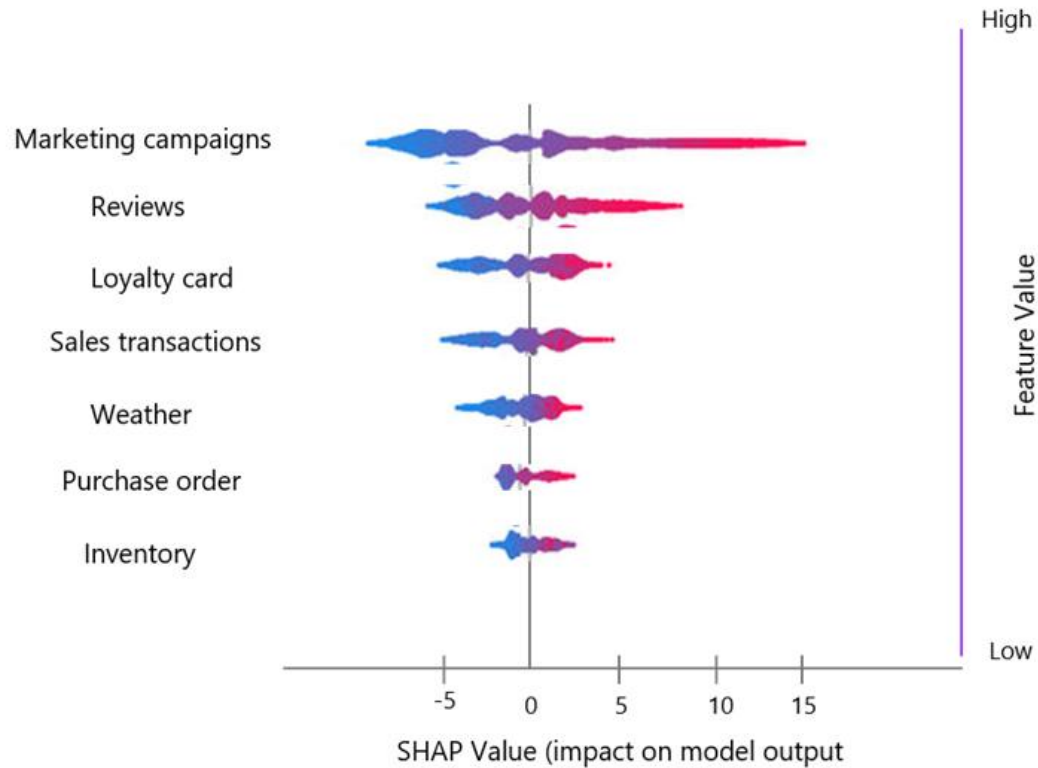
Figure 3: Toy example to present intuition for LIME. The black-box model's complex decision function f (unknown to LIME) is represented by the blue/pink background, which cannot be approximated well by a linear model. The bold red cross is the instance being explained. LIME samples instances, gets predictions using f , and weighs them by the proximity to the instance being explained (represented here by size). The dashed line is the learned explanation that is locally (but not globally) faithful.

Feature	Value
alcohol	9.50
sulphates	0.48
total sulfur dioxide	102.00
volatile acidity	0.50
density	1.00
citric acid	0.17
pH	3.39
free sulfur dioxide	21.00
residual sugar	1.60

Prediction probabilities



VISUALIZING LIME vs SHAP



Decoding Predictions

Force Plots show how each feature "pushes" the prediction away from the baseline:

- **Red features:** Push probability UP.
- **Blue features:** Push probability DOWN.

This level of granularity transforms model outputs into **actionable customer interventions**.

Real-World Applications of Explainable AI



HEALTHCARE

AI enhances trust in diagnoses and treatments with transparent decision-making.



FINANCE

Provides clear explanations for credit scoring, fraud detection, and investments.



AUTONOMOUS VEHICLES

Ensures safety by explaining real-time decisions like obstacle detection and navigation.



HUMAN RESOURCES

Promotes fair hiring practices with bias detection and transparency.